Child & Women Safety Wearable — Complete System Flow

1. Project Overview

A wearable device (ESP32-based) that protects a child or woman while walking alone, by detecting suspicious physical contact, tracking location, alerting parents/police automatically, and broadcasting an SOS to nearby app users — without relying on the wearer being able to act.

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2. Hardware Components

Component	Purpose
ESP32 (with built-in BLE)	Main controller — runs all logic
Capacitive touch sensor / FSR	Detects physical contact + duration

Component	Purpose
Vibration motor (ERM/coin type)	Feedback to child
Push button	Manual stop / manual SOS
MEMS microphone (INMP441)	Detects voice command "stop" / distress sound
GPS module (NEO-6M)	Live location tracking
GSM module (SIM800L)	Sends SMS/calls without needing WiFi
Buzzer + Strobe LED	Physical deterrent, alerts people nearby directly
Battery + charging circuit	Power

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3. Touch Escalation Flow

This is the core safety logic — designed so the *system* escalates automatically, since a frightened child may not be able to act.

```
Touch detected on wearable
        ↓
Level 1: Vibrate for 1 second (simple
notify)
        ↓
Touch continues 3–5 sec?
        ↓ YES
Level 2: Vibration becomes continuous
(monitoring state)
        ↓
Child presses STOP button OR says "stop"?
   ↓ YES                          ↓ NO (10–
15 sec pass)
Alert cancelled              Level 3:
ESCALATE
(friend/relative case)              ↓
                          • Send live GPS
location to parent's phone
                          • Send SMS alert to
parent (GSM)
                          • Send SMS + GPS to
police via 112 (ERSS)
                          • Start BLE
broadcast to nearby app users
                          • Continue
vibration as ongoing warning
                          • (Optional)
Activate buzzer/strobe as deterrent
```

Why this works: the child only needs to act to *cancel* a false alarm (friend touching, relative, etc.). If they can't respond — because they're

scared, restrained, or distracted — the system assumes danger by default and escalates on its own.

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4. Home ↔ School Geofence Alert

1. Parent opens the companion app and marks the home-to-school route (a path with a tolerance radius, e.g. 50m).
2. This route data is synced to the wearable.
3. ESP32 continuously compares live GPS position to the saved route.
4. If the child's position goes outside the tolerance radius for a set time (e.g. 30 sec), the wearable automatically sends an SMS alert + current GPS location to the parent via GSM — no manual trigger needed.

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5. Multi-Recipient Alert System

Recipient	Method	Notes
Parent	GSM SMS + GPS location	Always works, no app needed on parent's side for this

Recipient	Method	Notes
Police	GSM SMS/call to **112 (ERSS)**	India's official emergency response system — auto-routes to nearest police unit based on location. A wearable cannot look up "nearest police station" on its own, so 112 is the realistic and correct integration
Nearby people	**BLE broadcast**	Only reaches phones that have the companion app installed and are scanning in the background — it is a broadcast, not a targeted message, so it cannot reach random strangers without the app

How BLE broadcast works

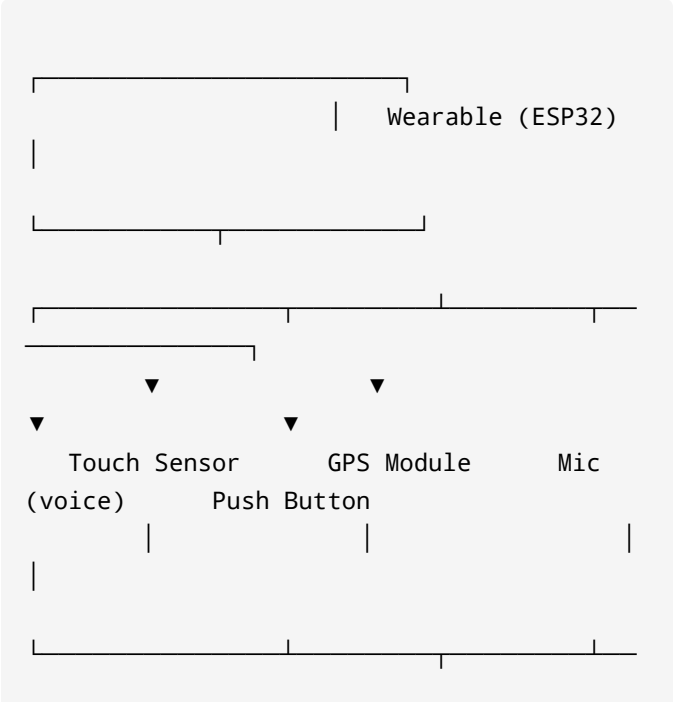
- On Level 3 escalation, ESP32 starts **BLE advertising**, repeatedly sending a small packet:

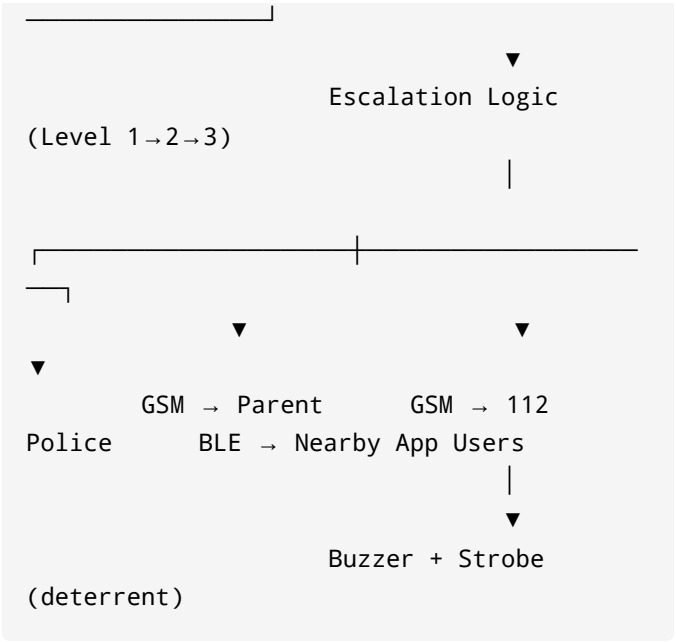
`SOS_ALERT | GPS coordinates | device ID`

- Any phone within range (~50 – 100m) running the companion app and scanning for this signal receives a push notification with a distinct alert sound and approximate distance/direction
- Phones without the app installed receive nothing — this is a hard technical limit of BLE, not a design gap

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6. Full System Flow (Summary)





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7. Full Flow in One Paragraph

When someone touches the child, the wearable gives a short 1-second vibration as a simple notification; if the touch continues for 3 – 5 seconds without stopping, the vibration becomes continuous, entering a monitoring state where the child can cancel it by pressing the stop button or saying "stop" — covering normal cases like friends or relatives touching them. If there is no response within 10 – 15 seconds, the system assumes danger and automatically escalates: it sends the

live GPS location and an SMS alert to the parent's phone via the GSM module, sends the same location and alert as an SMS/call to the police through India's 112 emergency system, and starts broadcasting a Bluetooth Low Energy SOS signal so that any nearby phone running the companion app is notified with a distinct alert and approximate distance, while the vibration keeps running and the buzzer/strobe LED can activate as a physical deterrent. Separately, the wearable also tracks the child's route from home to school using GPS; if the parent has set this path in the companion app and the child moves outside the allowed route for a set time, the wearable automatically sends a location alert to the parent, all without needing any manual action from the child.

8. Key Design Decisions (and why)

- **No gender detection** — not technically reliable and raises privacy concerns; dropped in favor of behavior-based detection (proximity + duration + no response).
- **Touch alone doesn't trigger full alert** — avoids false alarms in crowded areas; only sustained, unresolved contact escalates.
- **No "message any nearby phone number"** — telecom towers never expose nearby phone

numbers to any device; this is a privacy-protected barrier at the network level, not a hardware limitation. BLE broadcast to app users is the real, working alternative.

- **112 (ERSS) instead of custom "nearest police station" logic** — the wearable has no way to know police station locations; 112 already solves this at the government level.